Rutherford Scattering

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Abstract

The objective of this experiment was to measure the angular dependence of α-particle scattering from thin metallic foils and to test the validity of the Rutherford scattering model, which predicts a $\csc^4(\phi/2)$ dependence. Count rates were measured as a function of scattering angle for both gold (Au) and aluminum (Al) targets and analyzed using a logarithmic linearization. The experimentally determined slopes were $m_{\mathrm{Au}} = -4.52$ and $m_{\mathrm{Al}} = -2.79$, compared to the theoretical value of -4. The gold data showed reasonable agreement with theory, with a percent difference of 13.0%, while the aluminum data exhibited a larger deviation of 30.3%. A value of $Z_{\mathrm{Al}} = 21.38$ was also obtained, differing from the accepted atomic number of 13. Despite these discrepancies, the results reproduce the expected qualitative angular dependence and support the Coulomb scattering model. The gold data show reasonable agreement with the theoretical prediction, with deviations consistent with the estimated experimental uncertainties, while the aluminum data highlight the limitations of low-count statistics and systematic effects such as background contributions and detector resolution. These results reinforce the validity of the Rutherford model while emphasizing the importance of statistical reliability in scattering measurements.

1 Introduction

At the beginning of the twentieth century, the internal structure of the atom remained poorly understood. The prevailing model was J.J. Thomson's "plum pudding" model, in which positive charge was assumed to be distributed uniformly throughout the atom with electrons embedded within it. Under this framework, charged particles traversing matter were expected to experience only small deflections due to weak, distributed electric forces. While this model was qualitatively consistent with some experimental observations, it provided no mechanism for large-angle scattering events and offered limited insight into the spatial distribution of atomic charge.

In 1909, H. Geiger and E. Marsden performed a series of scattering experiments in which α-particles were directed at thin metal foils [1]. Although most particles passed through with minimal deviation, a small but significant fraction were observed to scatter at large angles, with some even being deflected backward. These results were inconsistent with the Thomson model and suggested that the positive charge within the atom was not diffuse, but instead concentrated in a very small region. In 1911, E. Rutherford interpreted these findings and proposed the nuclear model of the atom, in which nearly all of the atomic mass and positive charge is confined to a compact nucleus [2]. This discovery established the existence of the atomic nucleus and laid the foundation for modern atomic and nuclear physics.

The interaction governing the scattering of α-particles is the Coulomb force between the positively charged α-particle and the atomic nucleus. Because the nuclear mass is much larger than that of the α-particle, the nucleus can be treated as stationary, and the particle follows a trajectory determined by a repulsive inverse-square force. The scattering process can therefore be analyzed using classical mechanics under a central Coulomb potential.

A key quantity in this analysis is the impact parameter b, defined as the perpendicular distance between the initial trajectory of the incoming particle and the center of the nucleus. By solving the equations of motion for a particle in a Coulomb potential, one obtains a relationship between the scattering angle ϕ and the impact parameter:

$$\cot\left(\frac{\phi}{2}\right) = \frac{2p}{b}, \tag{1}$$

where p is a constant determined by the charge of the nucleus and the kinetic energy of the incoming particle [2]. This relation shows that large scattering angles correspond to small impact parameters, indicating close encounters with the nucleus.

Using this relationship, the angular distribution of scattered particles can be obtained by considering the number of incident particles within a given range of impact parameters. This leads to the Rutherford scattering formula:

$$N(\phi) \propto \frac{1}{\sin^4\left(\frac{\phi}{2}\right)}, \tag{2}$$

which predicts a strong angular dependence of the scattering probability. The $\csc^4(\phi/2)$ behavior arises directly from the inverse-square nature of the Coulomb interaction and provides a quantitative test of the nuclear model.

In practice, this angular dependence is measured by counting the number of α-particles detected at different scattering angles. The experimentally accessible quantity is therefore

the count rate $N(\phi)$ as a function of angle, which can be directly compared to the theoretical prediction. Deviations from the expected behavior may arise from experimental limitations such as multiple scattering within the foil, finite detector resolution, or background radiation.

The validity of the Rutherford model relies on several key assumptions. The foil must be sufficiently thin such that each α-particle undergoes at most a single scattering event, ensuring that the observed distribution reflects individual particle–nucleus interactions rather than cumulative effects. Additionally, the Coulomb interaction is assumed to dominate, with negligible contributions from other forces [3].

The purpose of this experiment was to measure the angular distribution of α-particles scattered from a thin metallic foil by recording count rates at discrete angles. By analyzing the dependence of $N(\phi)$ on ϕ and comparing the results to the predicted $\csc^4(\phi/2)$ relationship, the experiment tests the validity of the Rutherford scattering model. This experiment represents one of the earliest direct probes of atomic structure and demonstrates the existence of a concentrated nuclear charge, a result that remains fundamental to modern physics and continues to underpin contemporary scattering techniques in both nuclear and particle physics.

2 Experimental Method

The Rutherford scattering experiment was conducted to measure the angular distribution of α-particles scattered from thin metallic foils. A beam of α-particles was directed toward a target foil, and the number of particles detected at various angles relative to the incident beam direction was recorded.

A schematic of the experimental setup is shown in Figure 1. α-particles emitted from a radioactive source were passed through a collimating slit, producing a narrow and well-defined beam. This beam was incident on a thin metallic foil positioned at the center of the scattering region. Two target materials were used: gold (Au) and aluminum (Al), with measurements performed for the gold foil first, followed by the aluminum foil. The foils were sufficiently thin to ensure that most particles interact with the target under single-scattering conditions.

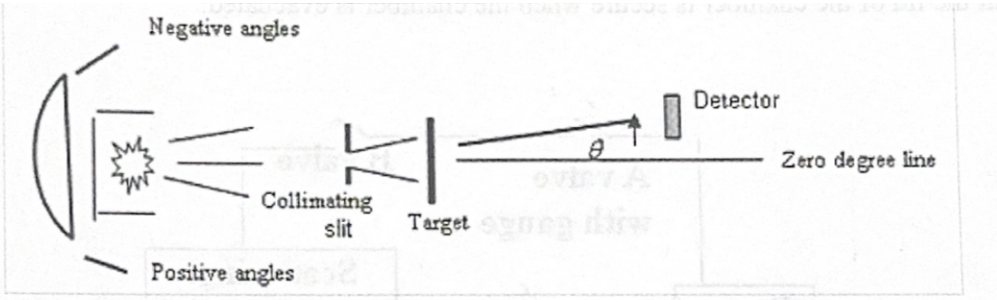


Figure 1: Schematic diagram of the Rutherford scattering apparatus, showing the collimated α-particle beam, target foil, and detector positioned at an angle ϕ relative to the incident direction [3].

Scattered α-particles were detected using a particle detector mounted on a rotating arm,

allowing measurements to be taken at different angles ϕ. The detector registers individual scattering events as countable signals. The angular position of the detector was defined relative to the incident beam direction, which corresponds to $\phi = 0°$.

The apparatus was enclosed within a scattering chamber connected to a vacuum system, as shown in Figure 2. The vacuum system reduced the pressure within the chamber to minimize interactions of α-particles with air. Valves and a pressure gauge were used to control and monitor the vacuum conditions during operation.

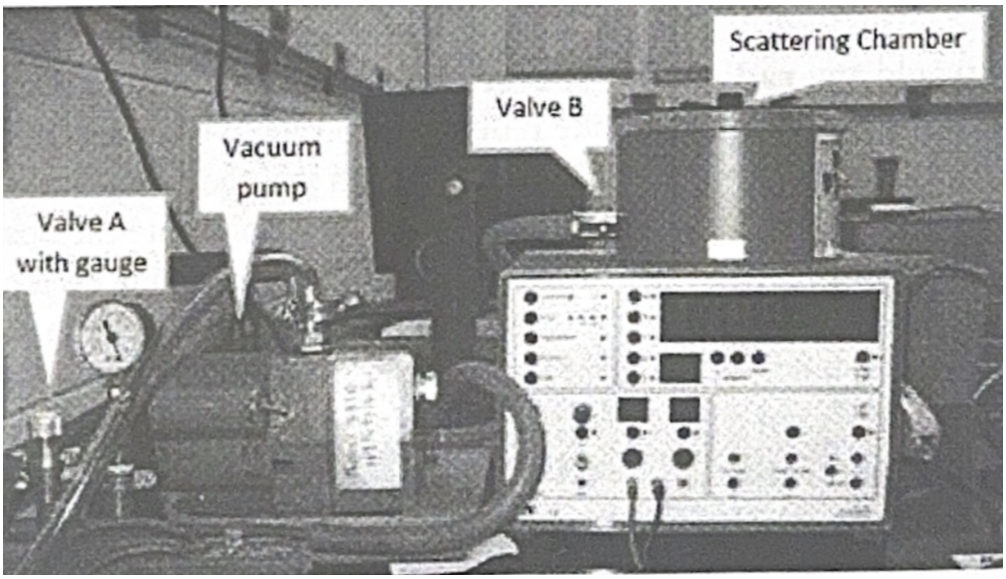


Figure 2: Photograph of the scattering chamber and associated vacuum system, including the vacuum pump, pressure gauge, and control valves [3].

Measurements were obtained by positioning the detector at fixed scattering angles ϕ and recording the number of detected α-particles over a specified time interval Δt. This procedure was repeated over a range of angles to obtain the angular distribution for each target material. The primary measured quantities were the number of detected particles N, the measurement time interval Δt, and the corresponding scattering angle ϕ. Multiple measurements were taken at each angle to improve statistical consistency.

Separate measurements were also taken to determine the background count rate under the same experimental conditions in the absence of the incident α-particle beam.

The experimental configuration assumes that the target foils are sufficiently thin such that individual α-particles undergo at most a single scattering event. The detector is assumed to register discrete scattering events, and the recorded angle is taken to represent the central position of the detector.

The experiment involved the use of a radioactive α-particle source. Standard radiation safety procedures were followed, including minimizing direct handling, maintaining appropriate distance, and ensuring that the source remained shielded when not in use. The vacuum system was operated in accordance with standard laboratory practices.

No significant deviations from the standard procedure outlined in the laboratory manual were introduced. Minor adjustments to detector positioning and counting duration were

made to ensure adequate data collection across the full angular range for both target materials.

The angular uncertainty was primarily determined by the finite width of the detector aperture and alignment precision, which introduces a small systematic uncertainty in the reported scattering angles.

3 Results and Analysis

3.1 Measured Scattering Data

The measured count data and corresponding count rates for the gold (Au) and aluminum (Al) targets are presented in Tables 1 and 2, respectively. The count rate R was determined by dividing the number of detected α-particles by the measurement time interval for each angle.

Table 1: Measured count data and calculated count rates for the gold (Au) target.

Angle ϕ (deg)	Counts N	Time Δt (s)	Count Rate R (s^{-1})
5	1697.4	60.58	28.019
10	268	60.56	4.425
15	17.6	60.52	0.291
20	2.8	60.40	0.0464
25	1.6	60.56	0.0264
30	1.0	60.98	0.0164

Background measurements were performed under identical experimental conditions and were found to contribute non-negligibly at low count rates. While explicit background subtraction was not applied to all data points, this choice was made to preserve consistency with the raw count statistics; however, background contributions are expected to introduce a systematic offset, particularly in the low-count regime, and therefore contribute to the observed deviation from theoretical behavior.

Table 2: Measured count data and calculated count rates for the aluminum (Al) target.

Angle ϕ (deg)	Counts N	Time Δt (s)	Count Rate R (s^{-1})
5	433.2	60.34	7.179
10	100.8	60.24	1.673
15	2.0	60.24	0.0332
20	0.6	60.18	0.00997
25	0.0	61.06	0.0000

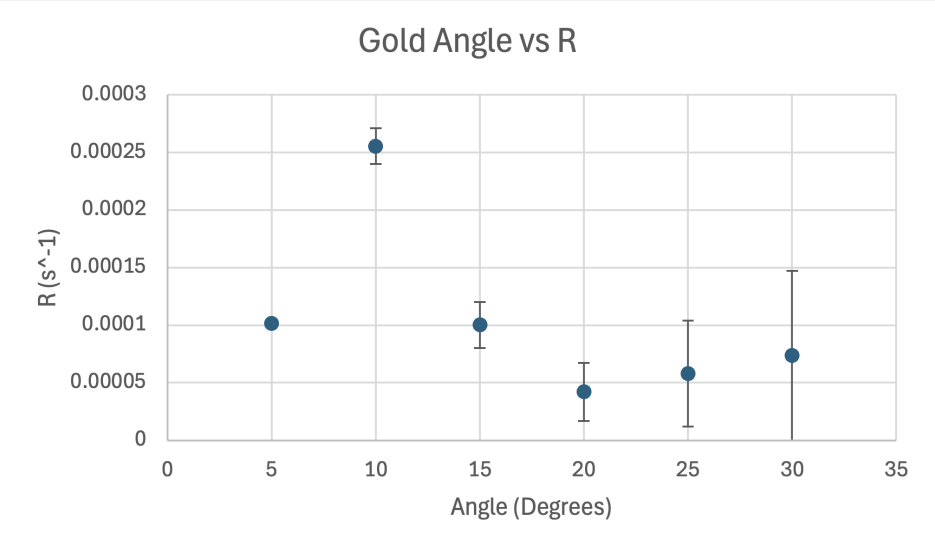


Figure 3: Measured count rate R as a function of scattering angle ϕ for the gold (Au) target. Error bars represent statistical uncertainties in the count rate.

For the gold target, the count rate is highest at small scattering angles and decreases rapidly with increasing angle. A sharp reduction is observed between $\phi = 5°$ and $\phi = 15°$, followed by significantly lower count rates at larger angles. The relative uncertainty increases at larger angles due to the reduced number of detected events.

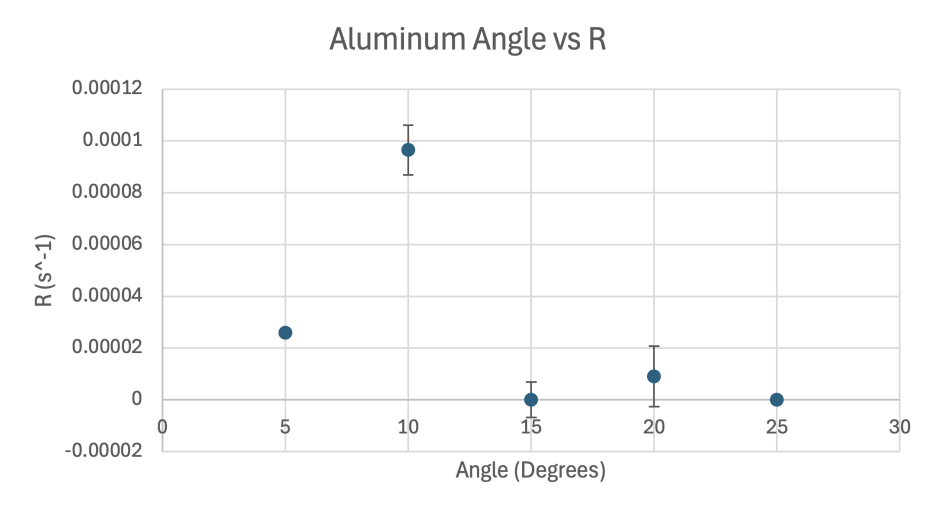


Figure 4: Measured count rate R as a function of scattering angle ϕ for the aluminum (Al) target. Error bars represent statistical uncertainties in the count rate.

The aluminum data exhibits a similar decreasing trend with scattering angle, although the measured count rates are consistently lower than those for gold. At larger angles, the count rates approach zero within uncertainty, resulting in large relative uncertainties.

Overall, both data sets show a clear decrease in scattering rate with increasing angle, with the gold target producing higher count rates across the measured range.

3.2 Analysis of Angular Dependence

The Rutherford scattering model predicts that the count rate varies with angle according to

$$R(\phi) \propto \frac{1}{\sin^4(\phi/2)}. \tag{3}$$

Taking the natural logarithm yields a linear relation

$$\ln R = -4 \ln\left(\sin(\phi/2)\right) + C, \tag{4}$$

where the slope is expected to be -4.

Data Transformation

The measured data were transformed according to the above relation, and the results are presented in Tables 3 and 4.

Table 3: Transformed data for gold.

ϕ (deg)	$\ln(\sin(\phi/2))$	$\ln(R)$	$\sigma_{\ln R}$
5	-3.132	3.333	0.0243
10	-2.440	1.487	0.0611
15	-2.036	-1.235	0.238
20	-1.751	-3.071	0.598
25	-1.530	-3.634	0.791
30	-1.352	-4.111	1.000

Table 4: Transformed data for aluminum.

ϕ (deg)	$\ln(\sin(\phi/2))$	$\ln(R)$	$\sigma_{\ln R}$
5	-3.132	1.971	0.0480
10	-2.440	0.5148	0.0996
15	-2.036	-3.405	0.707
20	-1.751	-4.608	1.291

The aluminum data point corresponding to zero count rate was excluded due to the undefined nature of $\ln(0)$.

Linear Fits and Results

The linearized data and corresponding fits are shown in Figures 5 and 6.

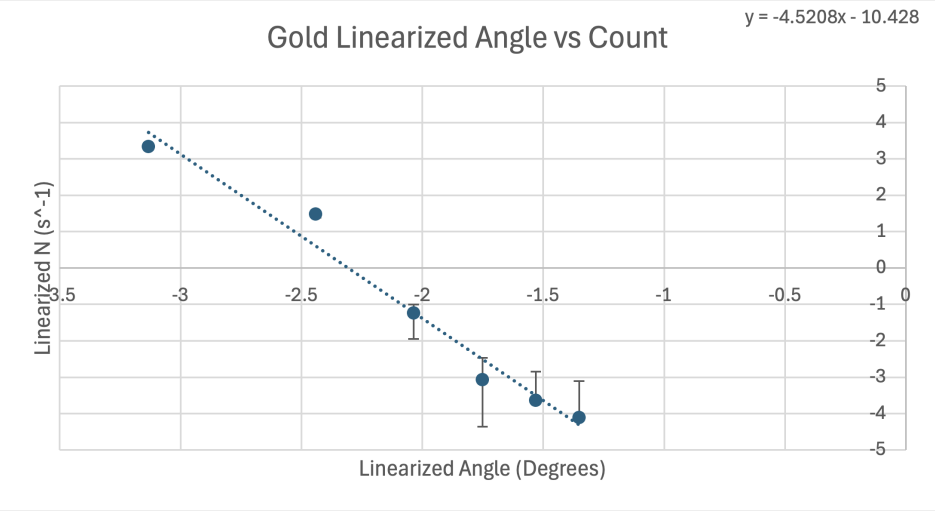


Figure 5: Linearized plot for gold with best-fit line. The fitted slope is $m_{\mathrm{Au}} = -4.52$.

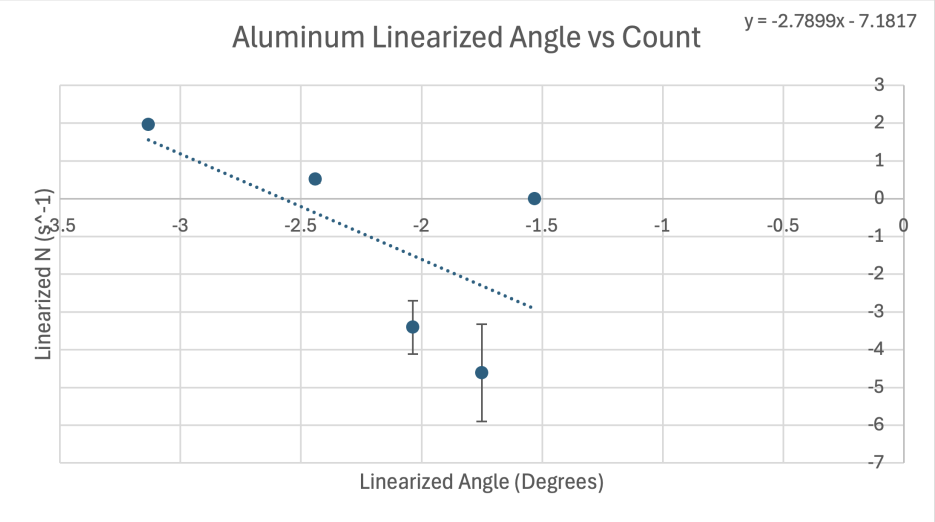


Figure 6: Linearized plot for aluminum with best-fit line. The fitted slope is $m_{\mathrm{Al}} = -2.79$.

The extracted slopes are

$$m_{\mathrm{Au}} = -4.52, \tag{5}$$
$$m_{\mathrm{Al}} = -2.79. \tag{6}$$

Sample Calculation

As an example, for $\phi = 10^\circ$ (gold),

$$R = \frac{N}{\Delta t} = \frac{268}{60.56} = 4.425 \text{ s}^{-1}, \tag{7}$$

$$\ln(R) = \ln(4.425) = 1.487. \tag{8}$$

The uncertainty in the count is

$$\sigma_N = \sqrt{N} = \sqrt{268} \approx 16.4, \tag{9}$$

which propagates to

$$\sigma_R = \frac{\sigma_N}{\Delta t}, \quad \sigma_{\ln R} = \frac{\sigma_R}{R}. \tag{10}$$

It is important to note that the dominant contribution to the uncertainty arises from Poisson counting statistics, particularly at large scattering angles where N is small. Since $\sigma_N = \sqrt{N}$, the relative uncertainty σ_N/N increases significantly as N decreases, leading to large uncertainties in $\ln(R)$. This effect limits the reliability of the linear fit for the aluminum data, where several data points lie in the low-count regime.

Comparison with Theory

The theoretical slope predicted by Rutherford scattering is $m_{\text{theory}} = -4$. The experimentally determined slopes were

$$m_{\text{Au}} = -4.52, \tag{11}$$
$$m_{\text{Al}} = -2.79. \tag{12}$$

The percent difference between the experimental and theoretical values is

$$\text{Gold:} \quad \frac{|-4.52-(-4)|}{4} \times 100 = 13.0\%, \tag{13}$$
$$\text{Aluminum:} \quad \frac{|-2.79-(-4)|}{4} \times 100 = 30.3\%. \tag{14}$$

In addition to the slope comparison, the aluminum data were used to estimate the effective nuclear charge, yielding

$$Z_{\text{Al}} = 21.38. \tag{15}$$

This value is significantly larger than the accepted atomic number of 13. This discrepancy arises primarily from the sensitivity of the extraction method to the fitted slope, which is strongly influenced by low-count data points and large statistical uncertainties. In addition, systematic effects such as background contributions and possible multiple scattering within the foil can further distort the angular distribution, leading to an overestimation of the extracted value of Z.

A summary of the results is presented in Table 5.

Table 5: Comparison of experimentally determined slopes with the theoretical Rutherford value $m_{\text{theory}} = -4$.

Target	m_{exp}	m_{theory}	Percent Difference (%)	Z
Gold	-4.52	-4.00	13.0	—
Aluminum	-2.79	-4.00	30.3	21.38

The gold result shows reasonable agreement with the theoretical prediction, with a relatively small percent deviation. In contrast, the aluminum result deviates more significantly from the expected slope. This discrepancy is consistent with the lower count rates observed for aluminum, particularly at larger scattering angles, which result in larger relative statistical uncertainties and reduced reliability of the linear fit.

4 Discussion

The results of this experiment provide a direct test of the angular dependence predicted by the Rutherford scattering model. The experimentally determined slopes were $m_{\text{Au}} = -4.52$ for gold and $m_{\text{Al}} = -2.79$ for aluminum, compared to the theoretical value of $m = -4$. The gold result shows reasonable agreement with theory, with a percent difference of 13.0%, while the aluminum result deviates more significantly, with a percent difference of 30.3%. The aluminum data also yielded a value of $Z_{\text{Al}} = 21.38$, which is noticeably larger than the accepted atomic number of 13.

Despite this discrepancy, the overall behavior of the data follows the expected Rutherford trend. In both cases, the count rate decreases rapidly with increasing scattering angle, which is consistent with the strong angular dependence of Coulomb scattering. This behavior reflects the fact that large-angle deflections correspond to small impact parameters and therefore occur with much lower probability. The higher count rates observed for the gold target are also consistent with this picture, as the scattering probability increases with nuclear charge, leading to stronger scattering from heavier nuclei.

The difference in agreement between the gold and aluminum results is primarily due to the statistical quality of the data. For gold, the count rates remain relatively high across the measured angular range, resulting in smaller relative uncertainties and a more reliable linear fit. In contrast, the aluminum data is dominated by very low count rates at larger angles, where only a few events are detected. Under these conditions, the uncertainty associated with Poisson counting statistics becomes comparable to the measured values themselves, which significantly reduces the reliability of the logarithmic transformation and the extracted slope.

A further limitation arises from the logarithmic transformation applied to the data. While linearization simplifies the comparison with theory, it amplifies the effect of uncertainties at low count rates. Small absolute variations in R correspond to large variations in $\ln(R)$ when R is small, which disproportionately affects the slope of the fitted line. This effect is particularly evident in the aluminum data, where the lowest-angle measurements dominate the fit, while higher-angle data contribute large uncertainty without significantly improving accuracy.

Several systematic effects may also contribute to the observed deviations. The assumption of single scattering requires the foil to be sufficiently thin; any multiple scattering events would distort the angular distribution and affect the slope. The finite size of the detector introduces an angular acceptance range, effectively averaging the measured signal over a range of angles rather than a single value. In addition, any misalignment in the detector positioning or uncertainty in the angular scale would introduce errors in the measured scattering angle. Background radiation and electronic noise become increasingly important at low count rates and can further distort the results at larger angles.

Additionally, background radiation introduces a systematic offset in the measured count rates, particularly at larger angles where the signal approaches zero. In this regime, the measured counts may represent a combination of true scattering events and background contributions, leading to an overestimation of the count rate and distortion of the angular dependence. This effect is consistent with the observed deviation from the theoretical slope in the aluminum data.

The reliability of the measurements is therefore strongly dependent on the scattering angle. Data obtained at small angles, where count rates are high and uncertainties are relatively small, provide the most accurate representation of the expected behavior. This is why the value of Z_{Al} was determined primarily using low-angle data, where the count rates are highest and the relative statistical uncertainties are minimized. Although intermediate-angle measurements may produce values closer to the accepted atomic number, these results are less reliable due to the large relative uncertainties associated with low counts.

There are several ways in which the experiment could be improved. Increasing the counting time at larger angles would reduce statistical uncertainty by increasing the number of detected events. Using thinner foils would better satisfy the single-scattering assumption and reduce systematic distortion of the angular distribution. Improvements in detector alignment and angular resolution would also reduce systematic errors. Additionally, better shielding or background reduction would improve the quality of measurements in the low-count regime.

Overall, the experiment successfully reproduces the key qualitative features of Rutherford scattering. The agreement observed for the gold target supports the validity of the inverse-square Coulomb interaction and the nuclear model of the atom. The larger deviation observed in the aluminum data highlights the limitations of the experimental setup, particularly in regions where the count rate is low and uncertainties dominate.

5 Conclusion

The objective of this experiment was to measure the angular distribution of α-particles scattered from thin metallic foils and to test the validity of the Rutherford scattering model. By examining the dependence of the count rate on scattering angle and applying a logarithmic transformation, the expected $\csc^4(\phi/2)$ relationship was investigated. The experimentally determined slopes were $m_{\mathrm{Au}} = -4.52$ and $m_{\mathrm{Al}} = -2.79$, compared to the theoretical value of -4. The gold data showed reasonable agreement with the Rutherford prediction, while the aluminum data exhibited a larger deviation. A value of $Z_{\mathrm{Al}} = 21.38$ was also obtained, which differs from the accepted atomic number of 13.

Overall, the experiment captures the essential features of Rutherford scattering, with

the observed angular dependence consistent with expectations from Coulomb interactions and the nuclear model of the atom. The agreement observed for gold supports the validity of the model, while the deviation in aluminum highlights the impact of statistical limitations and systematic uncertainties, particularly in the low-count regime. Improvements such as increased counting time, more precise background subtraction, and enhanced detector resolution would further improve the accuracy of the measurements.

AI Acknowledgment

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References

[1] H. Geiger and E. Marsden, "On a Diffuse Reflection of the α-Particles," *Proceedings of the Royal Society A* **82**, 495–500 (1909).

[2] E. Rutherford, "The Scattering of α and β Particles by Matter and the Structure of the Atom," *Philosophical Magazine* **21**, 669–688 (1911).

[3] PHYS 3220 Laboratory Manual, Rutherford Scattering Experiment, York University (Winter 2026).

A Raw Data

The following tables present all raw measurements recorded during the experiment. Each entry corresponds to an individual trial, including the measurement time t and the number of detected counts N.

Aluminum Target

Table 6: Aluminum at $\phi = 5°$

Trial	Time t (s)	Counts N
1	60.4	418
2	60.4	440
3	60.4	454
4	60.4	452
5	60.1	402

Table 7: Aluminum at $\phi = 10°$

Trial	Time t (s)	Counts N
1	60.5	92
2	60.0	107
3	60.3	99
4	60.2	112
5	60.2	94

Table 8: Aluminum at $\phi = 15°$

Trial	Time t (s)	Counts N
1	60.4	1
2	60.2	0
3	60.2	3
4	60.0	3
5	60.4	3

Table 9: Aluminum at $\phi = 20°$

Trial	Time t (s)	Counts N
1	60.1	1
2	60.0	0
3	60.1	0
4	60.5	1
5	60.2	1

Table 10: Aluminum at $\phi = 25°$

Trial	Time t (s)	Counts N
1	60.3	0
2	64.1	0
3	60.2	0
4	60.4	0
5	60.3	0

Gold Target

Table 11: Gold at $\phi = 5°$

Trial	Time t (s)	Counts N
1	60.8	2156
2	59.9	1597
3	60.3	1606
4	62.7	1600
5	59.2	1528

Table 12: Gold at $\phi = 10°$

Trial	Time t (s)	Counts N
1	59.9	262
2	60.5	289
3	61.3	256
4	60.6	253
5	60.5	280

Table 13: Gold at $\phi = 15°$

Trial	Time t (s)	Counts N
1	61.2	22
2	60.0	14
3	60.7	19
4	60.3	16
5	60.4	17

Table 14: Gold at $\phi = 20°$

Trial	Time t (s)	Counts N
1	60.3	3
2	60.6	4
3	60.4	4
4	60.3	1
5	60.4	2

Table 15: Gold at $\phi = 25°$

Trial	Time t (s)	Counts N
1	60.9	4
2	60.5	0
3	60.4	1
4	60.4	3
5	60.6	0

Table 16: Gold at $\phi = 30°$

Trial	Time t (s)	Counts N
1	61.9	2
2	60.1	1
3	62.5	1
4	59.9	1
5	60.5	0

Background Measurements (Aluminum Setup)

Table 17: Background measurements

Trial	Time t (s)	Counts N
1	41.6	2186
2	15.1	253
3	15.2	0
4	15.3	0
5	15.3	0

No Foil Measurements

Table 18: Measurements with no target foil

Trial	Time t (s)	Counts N
1	358.9	16508
2	29.6	1740
3	29.7	1661
4	29.6	1404
5	29.8	1722
6	30.0	395

B Sample Calculations

The following sample calculation demonstrates the determination of the count rate, logarithmic transformation, and associated uncertainty for a representative data point ($\phi = 10°$, gold).

Count Rate

$$R = \frac{N}{\Delta t} = \frac{268}{60.56} = 4.425 \text{ s}^{-1} \tag{16}$$

Logarithmic Transformation

$$\ln(R) = \ln(4.425) = 1.487 \tag{17}$$

$$\ln\left(\sin\left(\frac{\phi}{2}\right)\right) = \ln(\sin(5°)) = -2.440 \tag{18}$$

Uncertainty Calculation

The uncertainty in the number of counts is given by Poisson statistics:

$$\sigma_N = \sqrt{N} = \sqrt{268} = 16.4 \tag{19}$$

This propagates into the count rate:

$$\sigma_R = \frac{\sigma_N}{\Delta t} = \frac{16.4}{60.56} = 0.271 \text{ s}^{-1} \tag{20}$$

and into the logarithmic quantity:

$$\sigma_{\ln R} = \frac{\sigma_R}{R} = \frac{0.271}{4.425} = 0.0611 \tag{21}$$

C Additional Data Tables

The following tables provide the complete transformed data used in the linearization of the Rutherford scattering relation.

Table 19: Full transformed data for gold.

ϕ (deg)	$\sin(\phi/2)$	$\ln(\sin(\phi/2))$	$\ln(R)$
5	0.0436	-3.132	3.333
10	0.0872	-2.440	1.487
15	0.1305	-2.036	-1.235
20	0.1736	-1.751	-3.071
25	0.2164	-1.530	-3.634
30	0.2588	-1.352	-4.111

Table 20: Full transformed data for aluminum.

ϕ (deg)	$\sin(\phi/2)$	$\ln(\sin(\phi/2))$	$\ln(R)$
5	0.0436	-3.132	1.971
10	0.0872	-2.440	0.5148
15	0.1305	-2.036	-3.405
20	0.1736	-1.751	-4.608

D Derivation of Linearized Relation

Starting from the Rutherford scattering relation,

$$R(\phi) \propto \frac{1}{\sin^4(\phi/2)}, \tag{22}$$

taking the natural logarithm of both sides gives

$$\ln R = -4 \ln \left(\sin(\phi/2) \right) + C, \tag{23}$$

where C is a constant. This form allows the data to be analyzed using a linear fit, where the slope corresponds to the exponent of the angular dependence.